

MILESTONE 2

AI-Powered Workforce Analytics & Talent Intelligence Dashboard

Complete Integration & Documentation Report

REPORT BASIS	CONTENT
Assigned Task	Milestone 2 task supplied by the user
Team Inputs	Teammate-provided workflow, monitoring, validation, health-score and skill-gap documents
Purpose	Present the complete Milestone 2 scope, supporting work, results, limitations and integration

This report intentionally stays within the supplied Milestone 2 task and teammate documents. Where the documents identify a limitation, that limitation is retained.

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1. Milestone 2 Assigned Task

The Milestone 2 task assigned to the team contains four core responsibilities. These four points are the foundation for this report and define what must be implemented, validated and documented.

- **Develop AI analysis workflows for automated workforce trend monitoring and predictive insights.**
- **Implement machine learning models for attrition prediction and skill gap analysis.**
- **Create validation scripts for auditing prediction accuracy against historical workforce outcomes.**
- **Validate engine performance in generating reliable health scores and talent recommendations.**

My integration focus: connect the separate teammate outputs into one Milestone 2 workflow, preserve the documented validation evidence, clearly show what is implemented versus integration-ready, and prepare the final documentation/presentation.

2. Overall AI Analysis Workflow

The teammate workflow document defines Milestone 2 as a controlled process that starts with validated workforce data and continues through preprocessing, feature engineering, AI analysis, validation, insight generation, automation and dashboard-ready outputs. The stated AI branches are workforce trend monitoring, attrition prediction, workforce health/engagement analysis and skill-gap integration. ■filecite■turn6file1■L71-L82■

Stage	Role in the workflow
1. Workforce Dataset	Approved workforce records are provided as the input.
2. Data Quality Gate	Schema, target, identifiers, duplicates, missing/infinite values and data types are checked.
3. Preprocessing	Data is cleaned and transformed into model-ready form.
4. Feature Engineering	Target and usable model features are prepared.
5. Workforce Trends	Workforce KPIs and trend insights are generated.
6. Attrition Prediction	ML model produces prediction, probability and risk.
7. Validation	Predictions are audited against historical outcomes.
8. Health Analysis	Health/satisfaction indicators are evaluated against attrition.
9. Skill Gap	Skill-gap analysis is integrated when the required skills data exists.
10. Dashboard / Reports	Validated outputs are exposed as business-readable results.

3. Data Quality, Preprocessing & Control Gates

The workflow document places a data-quality gate before downstream AI processing. It checks the dataset structure and validates whether the data is suitable for the next stage. The documented workflow emphasizes that predictions should not be published before required quality and validation gates pass. ■filecite■turn6file1■L102-L134■

- Check the expected schema and target column.
- Check identifiers and duplicate records.
- Check missing and infinite values.
- Verify data types.
- Standardize columns/values and remove invalid records.
- Prepare the Attrition target and categorical variables for ML.
- Keep the process repeatable so the same preprocessing logic can be reused.

Control principle: Data-quality failure should stop downstream prediction, be logged and be notified rather than silently generating unreliable output.

4. Workforce Trend Monitoring

The supplied workforce trend-monitoring implementation calculates monthly workforce KPIs including total headcount, active count, turnover count, attrition rate, average tenure, average performance rating, total training hours and average training per employee. ■filecite■turn6file2■L210-L240■

- **Active Headcount:** current workforce size.
- **Attrition Rate:** workforce leaving during the reporting period.
- **Average Tenure:** average employee tenure.
- **Average Performance:** performance-level summary.
- **Training per Employee:** training exposure per employee.
- **Departmental Metrics:** comparison of workforce metrics by department.

The dashboard implementation supports KPI cards, trend visualization, reporting-period selection, departmental metrics and a configurable attrition alert threshold. ■filecite■turn6file6■L540-L587■

5. Automated Anomaly / Threshold Monitoring

The workforce monitoring component contains configurable threshold-based monitoring. In the documented validation example, a 5.0% attrition threshold was used. When the simulated attrition rate reached 6.0%, the system generated a CRITICAL alert. The validation suite reported that the pipeline tests passed.

Monitoring Item	Documented Example
Current attrition	6.0%
Configured threshold	5.0%
Result	CRITICAL alert
Reason	Attrition exceeded the configured threshold

This mechanism supports automated workforce trend monitoring by converting a KPI into a reviewable alert instead of requiring a user to manually inspect every period.

6. Attrition Prediction Model

The supplied attrition workflow uses a Random Forest classifier. The documented implementation uses the historical Attrition field as the target, converts Yes/No to binary form, prepares categorical variables, performs an 80/20 stratified split and produces probability and risk outputs. ■filecite■turn6file1■L139-L166■

Model Item	Documented Value
Algorithm	Random Forest Classifier
Number of trees	100
Class handling	Balanced
Training / testing	80% / 20%, stratified
Random state	42
Target	Attrition
Main outputs	Predicted class, attrition probability, Low/Medium/High risk

Purpose: identify employees who may have elevated attrition risk so that the result can be used as an analytical input for HR review and workforce planning.

7. Attrition Risk Outputs

The documented workflow converts model probability into a business-readable risk category. The output contract describes employee risk in terms of probability, risk category, prediction timestamp and model drivers.

Output	Meaning
Predicted Attrition	Model's predicted class.
Attrition Probability	Model probability associated with the predicted risk.
Risk Category	Low / Medium / High representation of the model output.
Prediction Timestamp	Time associated with the prediction output.
Top Model Drivers	Important model-derived factors used for interpretation.

8. Prediction Validation

The validation task audits prediction performance against historical workforce outcomes. The documented workflow uses 1,470 employee records and 34 columns, with Attrition as the target. It trains the Random Forest and compares predictions with historical outcomes. ■filecite■turn6file5■L434-L483■

Reported validation metrics

Metric	Reported Result
Accuracy	83.67%
Precision	44.44%
Recall	8.51%
F1 Score	14.29%

Correct interpretation: the reported 83.67% accuracy should not be treated as sufficient evidence of a highly reliable attrition model. The low recall and F1-score indicate that many actual attrition cases are missed. The teammate documentation therefore identifies further tuning and validation as necessary before autonomous production HR decisions. ■filecite■turn6file1■L169-L181■

9. Confusion Matrix & Employee-Level Validation

The validation workflow generates a confusion matrix and employee-level prediction information. The confusion matrix separates correct and incorrect prediction types and is useful for understanding where the model fails.

	Predicted No	Predicted Yes
Actual No	True Negative	False Positive
Actual Yes	False Negative	True Positive

Employee-level validation fields

- Employee ID
- Actual Attrition
- Predicted Attrition
- Attrition Probability
- Prediction Correct
- Risk Category

The teammate validation document describes Excel outputs for validation metrics, prediction comparison, confusion matrix, feature importance and complete employee-level validation results. ■filecite■turn6file5■L511-L556■

10. Feature Importance

The Random Forest workflow generates feature-importance information to help explain which available workforce variables contribute most strongly to the model's predictions.

Documented Important Features
Monthly Income
Age
Daily Rate
Total Working Years
Distance From Home
Monthly Rate
Years With Current Manager
OverTime

Important distinction: these are model-derived predictive features. They should not automatically be described as proven causal reasons for employee attrition.

11. Workforce Health Score Validation

The health-score validation report evaluates Job Satisfaction, Environment Satisfaction, Relationship Satisfaction and Work-Life Balance by comparing these indicators with the objective historical outcome of Attrition. The report states that the health metrics use standardized Likert-style ranges. ■filecite■turn6file9■L797-L845■

Work-Life Balance	Attrition Yes	Attrition No	Attrition Rate
1 (Bad)	10	6	62.5%
2 (Good)	4	21	16.0%
3 (Better)	5	82	5.7%
4 (Best)	1	35	2.8%

The supplied report identifies Work-Life Balance as an important indicator in the dataset and recommends prioritizing it, investigating low-satisfaction/retention profiles, and refining performance metrics. ■filecite■turn6file0■L47-L55■

Validation meaning: these results show an association in the supplied dataset. They should not be presented as universal HR rules or as proof of causation.

12. Skill Gap Analysis

The teammate skill-gap document proposes a Random Forest classification workflow using JobRole, ExistingSkills and RequiredSkills as inputs and GapStatus as the target. It defines accuracy, precision, recall and F1-score as evaluation metrics. Expected outputs include employee/job role, existing skills, required skills, missing skills, gap status, prediction and suggested training/upskilling area. ■filecite■turn6file7■L607-L693■

Required skill-gap flow

Step	Action
1	Collect employee/job-role information.
2	Identify existing skills.
3	Identify required role skills.

Step	Action
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4	Compare existing vs required skills.
5	Identify missing skills / gap status.
6	Generate training or upskilling area.
7	Validate and expose the result.

Dataset limitation: the current IBM HR Analytics Employee Attrition dataset does not contain an employee skills inventory or role-to-skill taxonomy. Therefore, actual employee-level skill-gap prediction cannot be claimed from that dataset alone. A separate skills dataset or defined role-to-skill mapping is required.

■filecite■turn6file7■L694-L710■

13. Automation, Controls & Output Contract

The workflow document defines triggers for new/updated datasets, scheduled refreshes, risk thresholds and retraining.

It also defines controlled exception handling so unreliable outputs are not silently published.

■filecite■turn6file4■L362-L387■

Condition	Required behavior
Data-quality failure	Stop downstream processing, log, notify and retain the last validated output.
Model-validation failure	Hold prediction publication, log and retain the last validated output.
Successful validation	Publish updated prediction and dashboard-ready output.
Threshold breach	Generate an alert for review.

Dashboard-ready output categories

- **Employee risk:** probability, risk category, prediction timestamp and top model drivers.
- **Model metrics:** accuracy, precision, recall, F1, confusion matrix and release status.
- **Trend metrics:** attrition rate, risk distribution and approved workforce dimensions.
- **Alerts:** high-risk count, threshold, severity and review status.
- **Recommendations:** validated business actions derived from approved upstream outputs.

14. How Teammate Work Maps to My Task

The following mapping shows how the documents supplied by the team support the four assigned Milestone 2 responsibilities.

My Assigned Task	Supporting Team Work	
Develop AI analysis workflows for automated workforce trend monitoring and predictive insights.	end m nitoring and predictive insights.	AI workflow design, workforce trend monitoring, dashboard and report implementation.
Implement ML models for attrition prediction and skill gap analysis.	analysis.	
Create validation scripts for auditing prediction accuracy against historical outcomes.	gainst historical outcomes.	Random Forest attrition workflow and proposed Random Forest skill-gap workflow.
Validate engine performance in generating reliable health scores and talent recommendations.	scores and talent recommendations.	

Validation workflow, model metrics, confusion matrix, prediction comparison and employee-level outcomes.

Health-score validation, documented findings/recommendations and dashboard-ready output contract.

15. Implementation / Integration Checklist

1. Confirm the common input dataset and data-quality gate.
2. Connect workforce trend-monitoring output to the dashboard/report layer.
3. Connect attrition model prediction and probability outputs.
4. Connect validation metrics and confusion matrix.
5. Connect employee-level validation results.
6. Connect feature-importance results.
7. Connect health-score analysis and documented recommendations.
8. Keep skill-gap module clearly marked as requiring skills data if that data is unavailable.

9. Apply threshold/alert logic.
10. Ensure validation failure does not publish unvalidated predictions.
11. Prepare final dashboard-ready outputs.
12. Prepare documentation and presentation evidence.

16. Final Presentation & Demo Flow

Slide	Recommended Content
1	Title – AI-Powered Workforce Analytics & Talent Intelligence
2	Milestone 2 assigned task
3	Overall AI workflow / architecture
4	Data quality and preprocessing
5	Workforce trend monitoring
6	Attrition prediction
7	Validation results
8	Health-score validation
9	Skill-gap analysis + dataset limitation
10	Integration of teammate outputs
11	Automation, controls and recommendations
12	Conclusion + live dashboard demo

Suggested live demo order

- Open the integrated dashboard.
- Show workforce KPI and trend monitoring.
- Show the attrition prediction/risk output.
- Open validation metrics and confusion matrix.
- Show employee-level prediction validation.
- Show feature importance.
- Show health-score findings and recommendations.
- Open the skill-gap section and explain the required skills-data limitation.
- Show the final integrated output and alert/validation status.

17. Important Limitations

- The supplied attrition model has strong reported accuracy but low recall and F1-score; therefore, it requires further tuning/validation.
- Health-score findings are associations in the supplied dataset and should not be treated as universal causal rules.
- The current IBM HR Attrition dataset does not contain the employee skills inventory or role-to-skill taxonomy required for actual skill-gap prediction.
- Skill-gap output should not be fabricated; the module requires appropriate skills data or a role-to-skill mapping.
- Model-derived feature importance should be described as predictive contribution, not causation.
- The final workflow should publish predictions only after required data-quality and model-validation gates pass.

These limitations are included because they are part of presenting the teammate work accurately and avoiding unsupported claims.

18. Conclusion

Milestone 2 brings together workforce trend monitoring, predictive attrition analysis, validation, health-score analysis and skill-gap integration into a controlled AI workflow. The teammate documents provide the individual technical components and evidence. The integration responsibility is to connect these outputs, preserve the validation results, document the workflow, expose the results through the final dashboard/report and clearly state where additional data or model improvement is required.

Final focus: implemented components + validated evidence + controlled integration + clear limitations + business-readable outputs.

Source Note

This report is based only on the Milestone 2 task supplied by the user and the teammate documents available in the conversation. Source-derived statements retain the terminology and results from those documents. No external research or unsupported employee-level skill information has been added.